

IO_PVG.h File Reference

IO Driver functions for PVG channels. [More...](#)

```
#include "IO_Driver.h"
```

Functions

IO_ErrorType **IO_PVG_Init** (**ubyte1** pvg_channel, **ubyte2** output_value)

Setup one PVG channel. [More...](#)

IO_ErrorType **IO_PVG_DeInit** (**ubyte1** pvg_channel)

Deinitializes one PVG output. [More...](#)

IO_ErrorType **IO_PVG_SetOutput** (**ubyte1** pvg_channel, **ubyte2** output_value)

Sets the output value of one PVG channel. [More...](#)

IO_ErrorType **IO_PVG_GetVoltage** (**ubyte1** pvg_channel, **ubyte2** *const voltage, **bool** *const fresh)

Get the voltage feedback of one PVG channel. [More...](#)

IO_ErrorType **IO_PVG_ResetProtection** (**ubyte1** pvg_channel, **ubyte1** *const reset_cnt)

Reset the output protection for a PVG channel. [More...](#)

Detailed Description

IO Driver functions for PVG channels.

Contains all service functions for the PVG (Proportional Valve Group) outputs. Up to 8 channels can be configured: [IO_PVG_00](#) .. [IO_PVG_07](#)

PVG valves are active valves with integrated electronic control. The input impedance is defined with 12kOhm to 50% Ubat (2 * 24kOhm to Ubat and GND).

The PVG output stage is a push/pull PWM output with a well defined output resistance of 2.58kOhm. This resistance is necessary for operation, low pass filtering and over load protection.

Note

The correction of the PVG impedance is directly performed inside the module. Therefore a PVG valve is needed for correct operation of this outputs.

When configuring a PVG output, the associated voltage feedback channel will also be configured.

PVG-API Usage:

- [Examples for PVG API functions](#)

PVG output protection

Each PVG output is individually protected against malfunction. Whenever the difference between the measured output (U_{feedback}) and the configured output calculated by $U_{\text{diff}} = \text{output_value} / 100 * U_{\text{BAT}} - U_{\text{feedback}}$ is greater than **abs(+/-18V)** for at least 100ms, the output protection is enabled latest within 12ms.

When entering the protection state, the PVG output has to remain in this state for at least 1s. After this wait time the PVG output can be reenabled via function `IO_PVG_ResetProtection()`. Note that the number of reenabling operations for a single PVG output is limited to 10. Refer to function `IO_PVG_ResetProtection()` for more information on how to reenable a PVG output.

PVG Code Examples

Examples for using the PVG API

PVG initialization example:

```
// Setup a PVG output with output value of 50%
IO_PVG_Init(IO_PVG_00,
            5000); // output value is 5000 = 50%
```

PVG task examples:

```
IO_PVG_SetOutput(IO_PVG_00,
                 6200); // set duty cycle to 62%
```

Function Documentation

```
IO_ErrorType IO_PVG_DeInit ( ubyte1 pvg_channel )
```

Deinitializes one PVG output.

Parameters

pvg_channel PVG output (**IO_PVG_00** .. **IO_PVG_07**)

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist
IO_E_CHANNEL_NOT_CONFIGURED	the given channel is not configured
IO_E_CH_CAPABILITY	the given channel is not a PVG channel
IO_E_UNKNOWN	an unknown error occurred

```
IO_ErrorType IO_PVG_GetVoltage ( ubyte1      pvg_channel,
                                ubyte2 *const voltage,
                                bool *const  fresh
                                )
```

Get the voltage feedback of one PVG channel.

Parameters

- pvg_channel** PVG channel (**IO_PVG_00** .. **IO_PVG_07**)
- [out] **voltage** Measured voltage in mV. Range: 0..32000 (0V..32.000V)
- [out] **fresh** Indicates if new values are available since the last call.
- **TRUE**: Value in "voltage" is valid
 - **FALSE**: No new value available.

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist
IO_E_NULL_POINTER	a NULL pointer has been passed to the function
IO_E_CHANNEL_NOT_CONFIGURED	the given channel is not configured
IO_E_CH_CAPABILITY	the given channel is not a PVG channel
IO_E_STARTUP	the output is in the startup phase
IO_E_REFERENCE	the reference voltage is out of range
IO_E_UNKNOWN	an unknown error occurred

Remarks

If there is no new measurement value available (for example the function **IO_PVG_GetVoltage()** gets called more frequently than the AD sampling) the flag **fresh** will be set to **FALSE**.

```
IO_ErrorType IO_PVG_Init ( ubyte1  pvg_channel,
                           ubyte2  output_value
                           )
```

Setup one PVG channel.

Parameters

pvg_channel PVG channel ([IO_PVG_00](#) .. [IO_PVG_07](#))

output_value Output value with which the PVG-channel will be initialized in percent * 100
(1000..9000)

Returns

[IO_ErrorType](#)

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist or is not available on the used ECU variant
IO_E_CH_CAPABILITY	the given channel is not a PVG channel or the used ECU variant does not support this function
IO_E_INVALID_PARAMETER	parameter is out of range
IO_E_CHANNEL_BUSY	the PVG output channel or the voltage feedback channel is currently used by another function
IO_E_DRIVER_NOT_INITIALIZED	the common driver init function has not been called before
IO_E_FPGA_NOT_INITIALIZED	the FPGA has not been initialized
IO_E_UNKNOWN	an unknown error occurred

```
IO_ErrorType IO_PVG_ResetProtection ( ubyte1          pvg_channel,
                                       ubyte1 *const reset_cnt
                                       )
```

Reset the output protection for a PVG channel.

Parameters

pvg_channel PVG channel ([IO_PVG_00](#) .. [IO_PVG_07](#))

[out] **reset_cnt** Protection reset counter. Indicates how often the application already reset the protection.

Returns

[IO_ErrorType](#)

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist
IO_E_FET_PROT_NOT_ACTIVE	no output FET protection is active
IO_E_FET_PROT_PERMANENT	the output FET is protected permanently because it was already reset more than 10 times
IO_E_FET_PROT_WAIT	the output FET protection can not be reset, as the wait time of 1s is not already passed
IO_E_CHANNEL_NOT_CONFIGURED	the given channel is not configured
IO_E_CH_CAPABILITY	the given channel is not a PVG channel
IO_E_UNKNOWN	an unknown error occurred

Attention

The protection can be reset 10 times, afterwards the output will remain permanently protected

Note

- After entering the output protection, a certain time has to pass before the output protection can be reset:
 - 1s for [IO_PVG_00](#) .. [IO_PVG_07](#)
- This function will not set the output back again to the last setpoint. After calling this function the output has to be set to the intended level with [IO_PVG_SetOutput\(\)](#)

Remarks

If the parameter `reset_cnt` is **NULL**, the parameter is ignored. The parameter `reset_cnt` returns the number of resets already performed.

```
IO_ErrorType IO_PVG_SetOutput ( ubyte1 pvg_channel,
                                ubyte2 output_value
                                )
```

Sets the output value of one PVG channel.

Parameters

pvg_channel PVG channel ([IO_PVG_00](#) .. [IO_PVG_07](#))

output_value Output value in percent * 100 (1000..9000)

Returns

[IO_ErrorType](#)

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist
IO_E_FET_PROT_ACTIVE	the output FET protection is active and has set the output to 0%. It can be reset with IO_PVG_ResetProtection() after the wait time is passed
IO_E_FET_PROT_REENABLE	the output FET protection is ready to be reset with IO_PVG_ResetProtection()
IO_E_FET_PROT_PERMANENT	the output FET is protected permanently because it was already reset more than 10 times. The output will remain at 0%
IO_E_INVALID_PARAMETER	the setpoint is out of range
IO_E_CHANNEL_NOT_CONFIGURED	the given channel is not configured
IO_E_CH_CAPABILITY	the given channel is not a PVG channel
IO_E_UNKNOWN	an unknown error occurred